Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

If $(1+x)\left(1+x+x^2\right)\left(1+x+x^2+x^3\right)\ldots(1+x+x^2+\ldots+x^n)=a_0+a_1x+a_2x^2+\ldots+a_mx^m$ then the value of a_1 is

(A) $m+1$

(B) $(n+1)$

(C) n

(D) m

Q.2

If the number of terms in the expansion of $\left(1-\frac{2}{x}+\frac{4}{x^2}\right)^n, x\neq 0$, is 28, then the sum of the coefficients of all the terms in this expansion is:

(A) 64

(B) 2187

(C) 243

(D) 729

Q.3

If $\left(1+x+x^2\right)^{48}=a_0+a_1x+a_2x^2+\ldots+a_{96}x^{96}$, then value of $a_0-a_2+a_4-a_6+\ldots+a_{96}$ is

(A) -1

(B) 0

(C) 1

(D) 48

Q.4

The coefficient of x^n in the expansion of $\left(1+\frac{1}{1!}x+\frac{1}{2!}x^2\cdot\cdot+\frac{1}{n!}x^n\right)^2$

(A) $\frac{2^n}{n!}$

(B) $\frac{2^n}{n}$

(C) $n!$

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(D) $\frac{1}{n!}$

Q.5

The value of the term independent of x in the expansion of $\left(1+\frac{x}{2}-\frac{2}{x}\right)^4, x \neq 0$ is equal to

(A) 1

(B) -6

(C) -5

(D) 6

Q.6

Coefficient of x^{25} in expansion of expression $\sum_{r=0}^{50} {}^{50}C_r(2x-3)^r(2-x)^{50-r}$ is

(A) ${}^{50}\mathrm{C}_{25}$

(B) $-{}^{50}\mathrm{C}_{30}$

(C) ${}^{50}\mathrm{C}_{30}$

(D) $-{}^{50}\mathrm{C}_{25}$

Q.7

Number of ways of selection of 8 letters from 24 letters of which 8 are a, 8 are b and the rest unlike, is given by

(A) 2^7

(B) 8.2^8

(C) 10.2^7

(D) None of these

Q.8

If 'n' is positive integer and three consecutive coefficient in the expansion of $(1+x)^n$ are in the ratio $6:33:110$, then n is equal to:

(A) 9

(B) 6

(C) 12

(D) 16

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Q.9

$f(1+x)^n = \sum_{r=0}^{n} C_r x^r$, then the value of
$C_0 + (C_0 + C_1) + (C_0 + C_1 + C_2) + \ldots\ldots$
$+ (C_0 + C_1 + C_2 + \ldots + C_n)$ is

(A) $n \cdot 2^{n-1}$

(B) $(n+2) \cdot 2^n$

(C) 2^n

(D) $(n+2) \cdot 2^{n-1}$

Q.10

If $C_0, C_1, C_2, \ldots\ldots, C_n$ be the coefficients in the expansion of $(1+x)^n$, then $\frac{2^2 \cdot C_0}{1.2} + \frac{2^3 \cdot C_1}{2.3} + \ldots. + \frac{2^{n+2} \cdot C_n}{(n+1)(n+2)}$ is equal to

(A) $\frac{3^{n+1}-2n-5}{(n+1)(n+2)}$

(B) $\frac{3^{n+2}-2n-5}{(n+1)(n+2)}$

(C) $\frac{3^{n+2}+2n-5}{(n+1)(n+2)}$

(D) None of these

Q.11

The coefficient of x^n in the expansion of $\left(1-9x+20x^2\right)^{-1}$ is

(A) $5^n - 4^n$

(B) $5^{n+1} - 4^{n+1}$

(C) $5^{n-1} - 4^{n-1}$

(D) None of these

Q.12

The coefficient of x^{53} in the expansion $\sum_{m=0}^{100} {}^{100}C_m (x-3)^{100-m} 2^m$ is

(A) ${}^{100}C_{47}$

(B) ${}^{100}C_{53}$

(C) $-{}^{100}C_{53}$

(D) $-{}^{100}C_{10}$

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Q.13

The coefficient of a^3b^4c in the expansion of $(1+a-b+c)^9$ is equal to

(A) $\frac{9!}{3!6!}$

(B) $\frac{9!}{4!5!}$

(C) $\frac{9!}{3!5!}$

(D) $\frac{9!}{3!4!}$

Q.14

If π (n) denotes product of all binomia coefficients in $(1+x)^n$, then ratio of $\pi(2002)$ to $\pi(2001)$ is

(A) 2002

(B) $\frac{(2002)^{2001}}{(2001)!}$

(C) $\frac{(2001)^{2002}}{(2002)!}$

(D) 2001

Q.15

If number of terms in the expansion of $(x-2y+3z)^n$ is 45, then $n=$

(A) 7

(B) 8

(C) 9

(D) 6^{10}

Answers & Solutions

Q.1 Answer:

n

Solution:

$a_1 =$ **coefficient of** x **in**
$(1+x)\left(1+x+x^2\right)\left(1+x+x^2+x^3\right)$
$\ldots\left(1+x+x^2+\ldots+x^n\right)$
$=$ **coefficient of** x **in** $(1+x)^n$
$= {}^nC_1 = n$

Q.2 Answer:

729

Solution:

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No. of terms in expansion of $\left(1-\frac{2}{x}+\frac{4}{x^2}\right)^{n}$ is

$$n^{n+3-1}c_{3-1}=28$$

$$n^{n+2}c_2=28$$

$$\frac{n+2!}{2!n!}=28$$

$$(n+2)(n+1)=8\times 7$$

$$n+2=8$$

$$n=6$$

$$\left(1-\frac{2}{x}+\frac{4}{x^2}\right)^{6}=(1-2+4)^6$$

$$=(3)^6=729$$

Q.3 Answer:

1

Solution:

Putting $x=i$, we get $\left(1+i+i^2\right)^{48}=a_0+a_1i+a_2i^2+\ldots+a^{98}i^{96}$

$$\Rightarrow \quad i^{48}=\left(a_0-a_2+a_4-\ldots\right)+i\left(a_1-a_3+\ldots\right)$$

Equating the real parts we get

$$a_0-a_2+a_4-a_6+\ldots+a_{96}=1$$

Q.4 Answer:

$\frac{2^n}{n!}$

Solution:

Coefficient of x^n in $\left(1+\frac{1}{1!}x+\frac{1}{2!}x^2+\ldots+\frac{1}{n!}x^n\right)^2$

$=$ **coefficient of x^n is** $\left(1+\frac{1}{1!}x+\frac{1}{2!}x^2+\ldots+\frac{1}{n!}x^n+\frac{1}{(n+1)!}x^{n+1}\ldots\right)^2$

$=$ **coefficient of x^n in** $e^{2x}=\left(1+2x+\frac{2^2x^2}{2!}+\frac{2^3x^3}{3!}\ldots\right)$

$=\frac{2^n}{n!}$

Q.5 Answer:

-5

Solution:

$$\left(1+\frac{x}{2}-\frac{2}{x}\right)^4$$

$$={}^4C_0+{}^4C_1\left(\frac{x}{2}-\frac{2}{x}\right)+{}^4C_2\left(\frac{x}{2}-\frac{2}{x}\right)^2$$

$$\quad+{}^4C_3\left(\frac{x}{2}-\frac{2}{x}\right)^3+{}^4C_4\left(\frac{x}{2}-\frac{2}{x}\right)^4$$

$$={}^4C_0+{}^4C_1\left(\frac{x}{2}-\frac{2}{x}\right)+{}^4C_2\left[\frac{x^2}{4}-2+\frac{4}{x^2}\right]$$

$$+{}^4C_3\left[{}^3C_0\left(\frac{x}{2}\right)^3-{}^3C_1\left(\frac{x}{2}\right)^2\left(\frac{2}{x}\right)+{}^3C_2\left(\frac{x}{2}\right)\left(\frac{2}{x}\right)^2-{}^3C_3\left(\frac{2}{x}\right)^3\right]$$

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$^4C_4\left[^4C_0\left(\frac{x}{2}\right)^4-{}^4C_1\left(\frac{x}{2}\right)^3\left(\frac{2}{x}\right)+{}^4C_2\left(\frac{x}{2}\right)^2\left(\frac{2}{x}\right)^2-{}^4C_3\left(\frac{x}{2}\right)\left(\frac{2}{x}\right)^3+{}^4C_4\left(\frac{2}{x}\right)^4\right]$

The term independent of x in above

$=^4C_0+{}^4C_2(-2)+{}^4C_4\cdot{}^4C_2=1-12+6=-5$

Q.6 Answer:

$-^{50}\mathrm{C}_{25}$

Solution:

$\sum_{r=0}^{50}{}^{50}C_r(2x-3)^r(2-x)^{50-r}$

$=[(2-x)+(2x-3)]^{50}$

$=(x-1)^{50}$

$=(1-x)^{50}$

$=^{50}\mathrm{C}_0-{}^{50}\mathrm{C}_1x\ldots\ldots-{}^{50}\mathrm{C}_{25}x^{25}+\ldots\ldots$

Coefficient of x^{25} is $-^{50}\mathrm{C}_{25}$

Q.7 Answer:

10.2^7

Solution:

The number of selection = coefficient of x^8 in

$\left(1+x+x^2+\ldots+x^8\right)\left(1+x+x^2+\ldots\ldots+x^8\right)\cdot(1+x)^8$

= coefficient of x^8 in $\frac{\left(1-x^9\right)^2}{(1-x)^2}(1+x)^8$

= coefficient of x^8 in $(1+x)^8$ in $\left(1+x^8\right)(1-x)^{-2}$

= coefficient of x^8 in

$\left({}^8\mathrm{C}_0+{}^8\mathrm{C}_1x+{}^8\mathrm{C}_2x^2+\ldots..+{}^8\mathrm{C}_8x^8\right)$

$\times\left(1+2x+3x^2+4x^3+\ldots..+9x^8+\ldots\right)$

$=9.{}^8\mathrm{C}_0+8{}^8\mathrm{C}_1+7.{}^8\mathrm{C}_2+\ldots.+1.{}^8\mathrm{C}_8$

$=\mathrm{C}_0+2\mathrm{C}_1+3\mathrm{C}_2+\ldots+9\mathrm{C}_8\left[\mathrm{C}_r={}^8\mathrm{C}_\mathrm{r}\right]$

Now $\mathrm{C}_0x+\mathrm{C}_1x^2+\ldots.+\mathrm{C}_8x^9=x(1+x)^8$

Differentiating with respect to x, we get

$\mathrm{C}_0+2\mathrm{C}_1x+3\mathrm{C}_2x^2+\ldots9\mathrm{C}_8x^8=(1+x)^8+8x(1+x)^7$

Putting $x=1$, we get

$C_0+2C_1+3C_2+\ldots.+9C_8=2^8+82^7=2^7(2+8)=10.2^7$

Q.8 Answer:

12

Solution:

Let the consecutive coefficient of

$(1+x)^{\mathrm{n}}$ **are** ${}^{\mathrm{n}}\mathrm{C}_{\mathrm{r}-1},{}^{\mathrm{n}}\mathrm{C}_{\mathrm{r}},{}^{\mathrm{n}}\mathrm{C}_{\mathrm{r}+1}$

From the given condition,

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$^{n}C_{r-1} : {}^{n}C_{r} : {}^{n}C_{r+1} = 6 : 33 : 110$

Now $^{n}C_{r-1} : {}^{n}C_{r} = 6 : 33$

$\Rightarrow \frac{n!}{(r-1)!(n-r+1)!} \times \frac{r!(n-r)!}{n!} = \frac{6}{33}$

$\Rightarrow \frac{r}{n-r+1} = \frac{2}{11} \Rightarrow 11r = 2n - 2r + 2$

$\Rightarrow 2n - 13r + 2 = 0$ **...(i)**

and $^{n}C_{r} : {}^{n}C_{r+1} = 33 : 110$

$\Rightarrow \frac{n!}{r!(n-r)!} \times \frac{(r+1)!(n-r-1)!}{n!} = \frac{33}{110} = \frac{3}{10}$

$\Rightarrow \frac{(r+1)}{n-r} = \frac{3}{10} \Rightarrow 3n - 13r - 10 = 0$ **...(ii)**

Solving (i) & (ii), we get $n = 12$

Q.9 Answer:

$(n+2) \cdot 2^{n-1}$

Solution:

We have

$S = C_0 + (C_0 + C_1) + (C_0 + C_1 + C_2) + \ldots + (C_0 + C_1 + \ldots + C_n)$

$= (C_0 + C_0 + \ldots n + 1 \text{times}) + (C_1 + C_1 + \ldots . n \text{ times }$

$(C_2 + C_2 + \ldots n - 1 \text{ times }) + \ldots .. + (C_{n-1} + C_{n-1}) + C_n$

$= (n+1)C_0 + nC_1 + (n-1)C_2 + \ldots . + 2C_{n-1} + C_n$

$= C_0 + 2C_1 + 3C_2 + \ldots .. + (n+1)C_n \quad [\because C_r = C_{n-r}]$

General Term $T_{r+1} = (r+1)C_r$

$T_{r+1} = r^{n}C_r + {}^{n}C_r = n \cdot {}^{n-1}C_{r-1} + {}^{n}C_r$

$\therefore S = \sum_{r=0}^{n} T_{r+1} = n \left[{}^{n-1}C_0 + {}^{n-1}C_1 + \ldots .. + {}^{n-1}C_{n-1}\right]$

$+ \left[{}^{n}C_0 + {}^{n}C_1 + \ldots .. + {}^{n}C_n\right.$

$= n.2^{n-1} + 2^n = (n+2)2^{n-1}$

Q.10 Answer:

$\frac{3^{n+2} - 2n - 5}{(n+1)(n+2)}$

Solution:

We have,

$$t_r + 1 = \frac{2^{r+2}{}^{n}C_r}{(r+1)(r+2)} = \frac{2^{r+2}}{r+2} \cdot \frac{1}{r+1}{}^{n}C_r$$

$$= \frac{2^{r+2}}{r+2} \cdot \frac{1}{n+1}{}^{n+1}C_{r+1}$$

$$= \frac{2^{r+2}}{n+1} \cdot \left(\frac{1}{r+2}{}^{n+1}C_{r+1}\right)$$

$$= \frac{2^{r+2}}{n+1} \cdot \frac{1}{n+2}{}^{n+2}C_{r+2}$$

$\left[\because \frac{1}{r+1}{}^{n}C_r = \frac{1}{n+1}{}^{n+1}C_{r+1}\right]$

Putting $r = 0, 1, 2, \ldots .., n$ **and adding we get,**

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The given expression

$= \frac{1}{(n+1)(n+2)} \left\{2^2 \cdot {}^{n+2}C_2 + 2^3 \cdot {}^{n+2}C_3 + \ldots \right.$
$\left. + 2^{n+2} \cdot {}^{n+2}C_{n+2}\right\}$
$= \frac{1}{(n+1)(n+2)} \left\{(1+2)^{n+2} - {}^{n+2}C_0 - 2 \cdot {}^{n+2}C_1\right\}$
$= \frac{3^{n+2}-2(n+2)-1}{(n+1)(n+2)} = \frac{3^{n+2}-2n-5}{(n+1)(n+2)}$

Q.11 Answer:

$5^{n+1} - 4^{n+1}$

Solution:

$$\left(1 - 9x + 20x^2\right)^{-1} = \left[(1-4x)(1-5x)\right]^{-1}$$
$$= \frac{1}{x}\left[\frac{(1-4x)-(1-5x)}{(1-4x)\cdot(1-5x)}\right] = \frac{1}{x}\left[(1-5x)^{-1} - (1-4x)^{-1}\right]$$
$$= \frac{1}{5}\left[(5-4)x + \left(5^2-4^2\right)x^2 + \left(5^3-4^3\right)x^3\right.$$
$$\left. + \ldots\ldots + \left(5^n - 4^n\right)x^n + \ldots\ldots\right]$$

coeff. of $x^n = 5^{n+1} - 4^{n+1}$

Q.12 Answer:

$-{}^{100}C_{53}$

Solution:

The given sigma is the expansion of

$[(x-3)+2]100 = (x-1)100 = (1-x)100$

Therefore, x^{53} **will occur in** T_{54}**.**

$T_{54} = {}^{100}C_{53}(-x)^{53}$

Therefore, the coefficient is $-{}^{100}C_{53}$

Q.13 Answer:

$\frac{9!}{3!4!}$

Solution:

$$(1 + a - b + c)^9$$
$$= \sum \frac{9!}{x_1!x_2!x_3!x_4!} \cdot (1)^{x_1}(a)^{x_2}(-b)^{x_3}(c)^{x_4}$$
$$\Rightarrow \text{Coefficient of } a^3b^4c = \frac{9!}{1!3!4!1!} = \frac{9!}{3!4!}$$

Q.14 Answer:

$\frac{(2002)^{2001}}{(2001)!}$

Solution:

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$$\frac{\pi(n)}{\pi(n+1)} = \frac{{}^{n}C_0 \cdot {}^{n}C_1 \cdot {}^{n}C_2 \ldots {}^{n}C_n}{{}^{n+1}C_0 \cdot {}^{n+1}C_1 \cdot {}^{n+1}C_2 \ldots \cdots {}^{n+1}C_{n+1}}$$

$$= \frac{1}{n+1} C_0 \left(\frac{{}^{n}C_0}{{}^{n+1}C_1}\right)\left(\frac{{}^{n}C_1}{{}^{n+1}C_2}\right) \ldots \ldots \cdot \left(\frac{{}^{n}C_n}{{}^{n+1}C_{n+1}}\right)$$

$$= \frac{1}{1}\left(\frac{1}{n+1}\right)\left(\frac{2}{n+1}\right) \ldots \ldots \cdot \left(\frac{n+1}{n+1}\right)$$

$$\left[\because \frac{{}^{n}C_r}{n+1_{C+1}} = \frac{r+1}{n+1}\right]$$

$$= \frac{(n+1)!}{(n+1)^{n+1}} = \frac{n!}{(n+1)^n}$$

$$\therefore \frac{\pi(2002)}{\pi(2001)} = \frac{(2002)^{2001}}{(2001)!}$$

Q.15 Answer:

8

Solution:

No. of terms in the expansion $= {}^{n+3-1}C_{3-1}$

$\therefore {}^{n+2}C_2 = 45 \Rightarrow n = 8$

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